Supplementary Materials

Controlled out-of-season spawning of reef-forming corals using offset environmental cues

Lonidas P. Koukoumaftsis[1*], Matthew Salmon[1], Glenn Everson[1], David J. Hughes[1], Deepa Varkey[2], Andrew J. Heyward[2,3], Andrea Severati[1,2] and Craig Humphrey[1]

[1] The National Sea Simulator, Australian Institute of Marine Science, Townsville, 4810, Queensland Australia
[2] Australian Institute of Marine Science, Townsville, 4810, Queensland, Australia
[3] University of Western Australia, 39 Fairway Street, Crawley, 6009, WA, Australia

*Corresponding author: l.koukoumaftsis@aims.gov.au

Proceedings of the Royal Society B: 10.1098/rspb.2025.1558

This PDF file includes:

1. Supplementary Methods

1.1 Coral larvae settlement on plug trays
1.2 System setup for out-of-season spawning
1.3 Offset photoperiod terminology
1.4 F2 life cycle closure
1.5 Assessing percentage of colonies spawning
1.6 Assessing fertilisation success
1.7 Assessing settlement rate
1.8 Assessing oocyte (egg) diameter

2. Supplementary figures and tables

2.1 Custom made Poly Vinyl Chloride trays
2.2 Lighting and life support systems for the F1 coral holding system
2.3 Concentrations of dissolved nutrients and phosphate
2.4 Temperature setpoints applied during the out-of-season coral spawning study
2.5 Photosynthetically Active Radiation provided to out-of-season spawning systems
2.6 Moonlight illuminance (lux) provided to out-of-season spawning systems

2.7 Out-of-season spawning observations

2.8 Out-of-season larval culture fertilisation percentages

2.9 Percentage of in-season corals spawned from the literature

2.10 Fertilisation percentages for in-season corals from the literature

2.11 Settlement assays from out-of-season larval cultures

2.12 Out-of-season coral oocyte (egg) diameters

2.13 Physiological traits measured for out-of-season corals against conspecifics reported in the wider literature

2.14 Results of NAFM/TAS statistical analysis

3. Supplementary Results

3.1 Comparative analysis of coral reproductive performance between OSS and literature benchmarks

4. Descriptions of accompanying electronic supplementary data files

5. Supplementary References

1. Supplementary Methods

1.1 Coral larvae settlement on plug trays

Coral larval settlement was performed in 50 L acrylic tanks housing PVC plug trays – each holding 165 aragonite plugs (figure. S1), and seawater exchange was provided continuously at a flow rate of 0.2 L min-1. Approximately 1200 coral larvae were introduced into each 50 L acrylic tank, and once larvae were deemed sufficiently attached to the aragonite plugs (typically ~24 h), the seawater exchange rate was increased to 0.8 L min-1 (providing one system exchange per hour).

1.2 System setup for out-of-season spawning

Biological filtration capacity was provided by ceramic biological media (MarinePure®, CerMedia, Buffalo, NY, USA). Seawater temperature was continuously monitored by a PT100-type FEP-insulated resistance temperature detector (Omega, Deckenpfronn, Germany) interfaced to a PCS7 Siemens Supervisory Control and Data Acquisition (SCADA) control system. Dynamic control of aquarium temperature was achieved using logic feedback applied to electric actuating valves, allowing a variable percentage of cold (15°C) or hot (40°C) water to pass through a shell-and-tube heat exchanger system (figure S2) to achieve system temperature within ±0.3°C of the programmed setpoint. System water was continuously recirculated by a centrifugal magnetic drive pump (MX 400, Iwaki, Tokyo, Japan), providing flow of 40 L min^{-1} to each tank, with ~90 L min^{-1} passing through the heat exchanger (Figure S2). In-tank water circulation was provided by two gyre pumps (XF280, Maxspect, Hong Kong, China).

Levels of nitrate (NO_3^-) and phosphate (PO_4^{3-}) were measured weekly using a segmented flow analyser (Seal AA3, Norderstedt, Germany) and did not exceed 5 μM and 0.5 μM

respectively throughout the study except during periods of gamete release where in-tank nutrient levels were temporarily elevated (figure S3).

1.3 Offset photoperiod terminology

In nature, sunset refers to the time at which the sun disappears over the horizon which is followed by a short period of twilight (where light still exists in the sky) before darkness. The variable period of darkness between twilight and moonrise plays an important role in coral spawning [1] and thus ensuring aquarium lights turn off at the required time corresponding to last light is critical for captive coral spawning programs. Here, we programmed our aquarium LEDs to come on (**first light**) at a time corresponding to the start of pre-dawn Civil Twilight, and go off (**last light**) at a time corresponding to the end of post-sunset Civil Twilight (electronic supplementary material, *Data_3_Photoperiod_entire.csv*) for the 6-month future photoperiod at Townsville, Australia (https://www.timeanddate.com) Since commercial aquarium LED systems often use the terms "sunrise" and "sunset" to describe the time that lights go on or off (i.e., what is referred to here as first light and last light), it is important to verify terminology for any given lighting system in use. Importantly, our calculations of spawning time after sunset (TAS) are based on the time elapsed between the expected sunset time on the date of spawning (electronic supplementary material, *Data_3_Photoperiod_entire.csv*) and the release of egg/sperm bundles from a given individual coral colony.

1.4 F2 life cycle closure

To close the lifecycle of the coral species in the out-of-season spawning program, larvae were settled during May and June spawning events in both 2022 and 2023 following methodology as described in S1.1, Coral larvae settlement on plug trays. Briefly, second filial generation (F2) coral recruits were held in 50 L aquaria supplied with filtered (1 µm) seawater providing

one system turnover per hour. F2 coral recruits were inoculated seven days post-settlement with cultured photosynthetic endosymbionts (Family: Symbiodiniaceae, [2]) specifically: *Cladocopium proliferum* [3] and *Durusdinium cf. trenchii* [2] at densities of 3000 and 5000 cells mL^{-1} respectively. Illumination over each tank was provided by a single LED panel (Hydra® 52, Aqua Illumination®, USA). Upon settlement, a photon irradiance (400 – 700 nm) of 15 µmol photons m^{-2} s^{-1} was provided for the first 14 days and slowly increased to 150 µmol photons m^{-2} s^{-1} over three months. Recruits were fed once daily with enriched rotifers at a final density of 0.5 rotifers mL^{-1}, and after one month instar I *Artemia salina* was also provided once daily at a final density of 0.5 nauplii mL^{-1}.

1.5 Assessing percentage of colonies spawning

To benchmark out-of-season spawning success, we compared the percentage of colonies that spawned in our study to values reported for captive GBR corals maintained under ambient (in-season) environmental profiles. We first extracted relevant data from cited literature involving F0 or F1 corals [4–6], then expanded this dataset by conducting a Google Scholar search on 25 April 2025 using the following terms: "colonies spawned" OR "spawned colonies" OR "corals spawned" OR "spawned corals" OR "coral spawned" OR "spawned coral" OR "proportion spawned" AND "Great Barrier Reef" AND ("*Acropora millepora*" OR "*Acropora loripes*" OR "*Acropora elseyi*" OR "*Acropora austera*" OR "*Acropora hyacinthus*" OR "M*ontipora aequituberculata*"). Each paper was manually screened and included if it met the following criteria: i) the work involved GBR corals, ii) colonies were not specifically selected for signs of gravidity iii) a minimum sample of five colonies were assessed, and iv) the proportion of spawning colonies was explicitly reported. A total of seven observations across three coral species met these criteria (table S3) with literature ranges and averages presented in table S7.

1.6 Assessing fertilisation success

To compare out-of-season fertilisation success to that of natural (wild) corals, we conducted a Google Scholar search on 28 April 2025) using the following terms: "fertilisation success" OR "fertilization success" AND "Acropora millepora" OR "Acropora loripes" OR "Acropora elseyi" OR "Acropora austera" OR "Acropora hyacinthus" OR "Montipora aequituberculata" AND ("Great Barrier Reef" AND "Great Barrier Reef Marine Park Authority" OR "GBRMPA"). The latter search terms were included to restrict results to studies using corals collected from the GBR region (where a GBRMPA permit is typically required). We manually screened studies across all years and included those reporting fertilisation percentages assessed under laboratory conditions using GBR-collected corals. Data were only included if fertilisation attempts were conducted under ambient temperatures and salinities, excluding experiments involving stressors such as elevated temperature, altered pH , anomalous salinity or contaminants).

This initial search yielded 22 observations from 11 studies: 20 values for *Acropora millepora*, and one each for *Acropora hyacinthus* and *Acropora elseyi*. Due to the limited data for species other than *A. millepora*, we conducted a second search on 29 April 2025 using the same search terms but substituting the term “collected” in place of “GBRMPA” or "Great Barrier Reef Marine Park Authority" to capture studies that reported coral origin differently. Studies were manually screened as described above and produced 9 additional observations across 6 studies. In total, 31 fertilisation observations from 17 studies across five coral species met our inclusion criteria (table S4) with ranges and averages summarised in table S7. No suitable data were found for *A. austera*.

1.7 Assessing settlement rate

Larval settlement assays were performed on a subset of cultures produced between May 2022 and June 2023 (table S5). For each assay, two six-well plates were maintained in a temperature-controlled room. Each well was filled with 10 mL of ultrafiltered (1 µm) seawater and approximately 10 coral larvae were transferred into each well using a 1.5ml sterile plastic pipette. Three wells per plate were treated with a chip of *Porolithon cf. onkodes* crustose coralline algae (CCA), while the remaining three served as controls without CCA.

Larvae were allowed to settle for 24 hours, after which the number of both settled and non-settled individuals were counted under a dissection microscope (Leica MZ6). Settlement was defined following Randall et al. [5], as firm larval attachment to the surface, with a pronounced flattening of the oral–aboral axis, the presence of early septal mesenteries radiating from a central mouth and deposition of a basal plate.

To benchmark out-of-season settlement performance, we compared our results to in-season settlement rates reported by Randal et al. [5] for five of the six species assessed in this study: *A. millepora, A. austera, A. hyacinthus, A. loripes, and M. aequituberculata*. We selected this dataset because i) broodstock were sourced from the same ecoregion, ii) identical assay protocols were used, and iii) experiments were performed under comparable laboratory conditions. Where Randal et al. [7] reported ”0” settlement for all replicates from early larval stages, these data points were excluded from average calculations (i.e., assuming that larvae had not yet reached competency). Values are reported as species-specific averages and ranges in table S7.

1.8 Assessing oocyte (egg) diameter

Egg diameters for out-of-season spawning events were measured from photographs taken after fertilisation attempts. Measurements of unfertilised eggs were performed using ImageJ, based on the mean of vertical and horizontal diameters for each oocyte (table S6). For each species, a minimum of n=25 egg samples were measured. To compare out-of-season egg sizes to natural spawning events, we used average oocyte diameter values from Randal et al. [7], which included raw measurements for natural (in-season) spawning across five of the six study species. The ranges reported in table S7 reflects these raw values and incorporate prior literature cited in Randall et al. [7] [8–11]. Data for *Acropora elseyi* were not published in Randall et al. [7]; therefore the values reported in supplementary table S7 are based on unpublished data from C. Randall (2017, AIMS) and supplemented with measurements from Wallace [8].

2. Supplementary figures and tables

2.1 Custom made Poly Vinyl Chloride trays

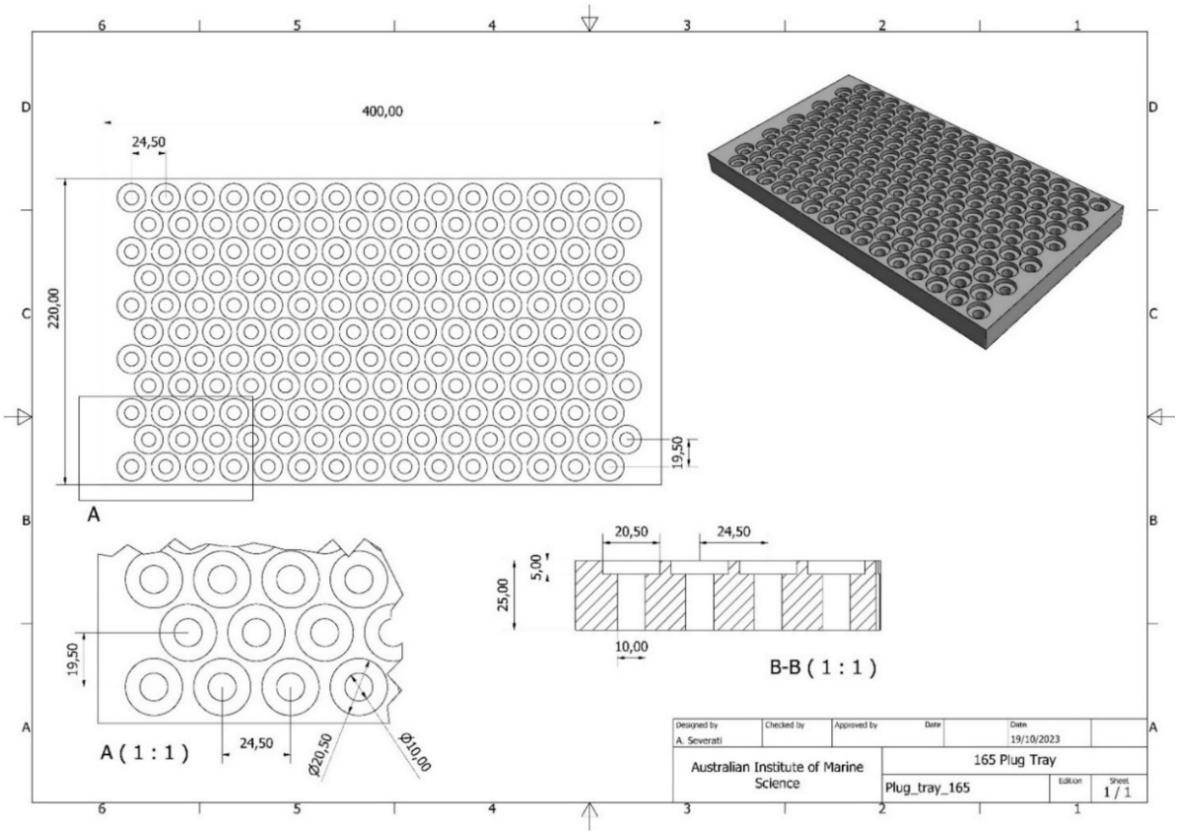


Figure S1. Custom-made Poly Vinyl Chloride (PVC) trays used to hold 165 Ocean Wonders® 20 mm aragonite plugs for coral larval settlement during the out-of-season spawning project.

2.2 Lighting and life support systems for the F1 coral holding system

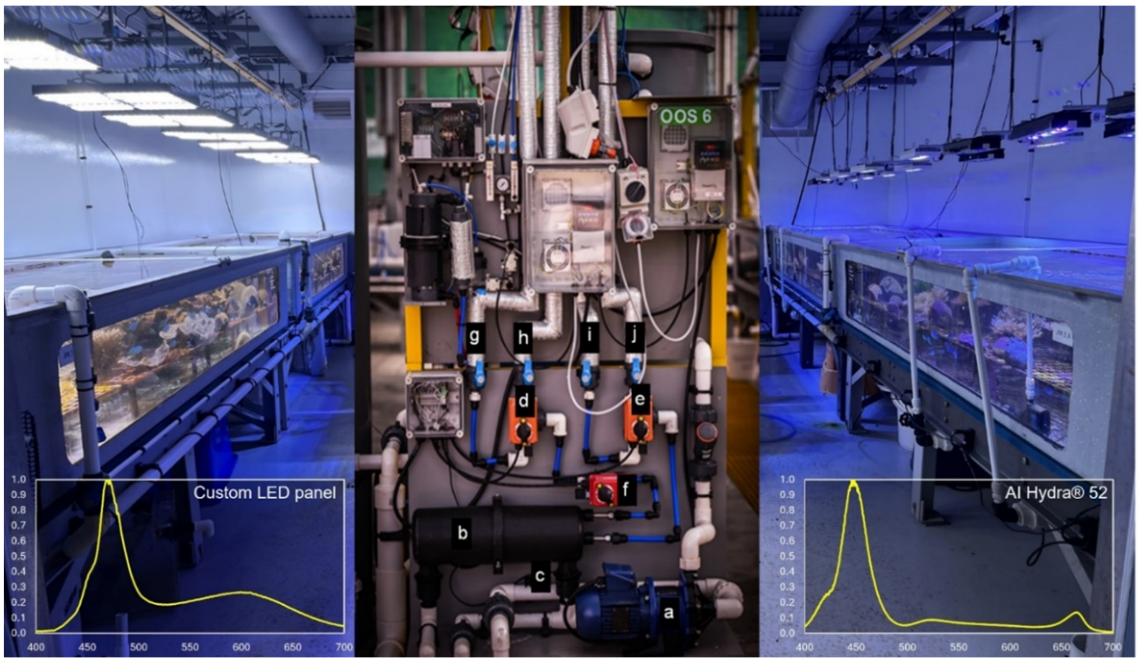


Figure S2. Lighting and life support system components of the F1 coral holding system. a) Metallic centrifugal magnetic drive pump b) Shell-and-tube heat exchanger c) Fluorinated ethylene propylene (FEP)-insulated resistance temperature detector (d–f) electric actuating valves for temperature regulation, (g–j) cold and hot water inlet and return lines. Inset panels show spectral outputs of the two LED units used (see Materials and Methods for details).

2.3 Concentrations of dissolved nutrients and phosphate

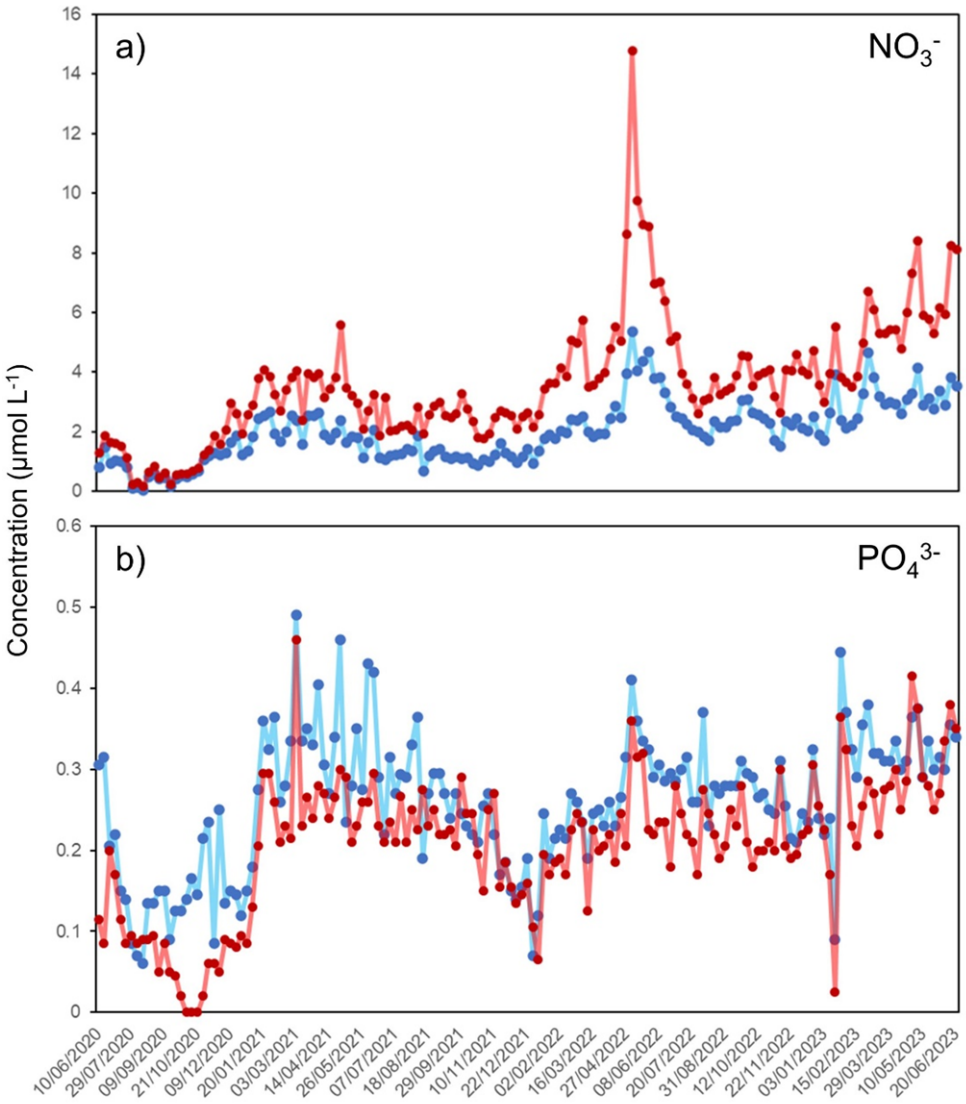


Figure S3. Temporal profiles of dissolved nutrient concentrations measured in out-of-season coral spawning holding systems during the study: a) nitrate (NO_3^-) and b) phosphate (PO_4^{3-}) . Blue points represent System 1 and red points represent System 2.

2.4 Temperature setpoints applied during the out-of-season coral spawning study

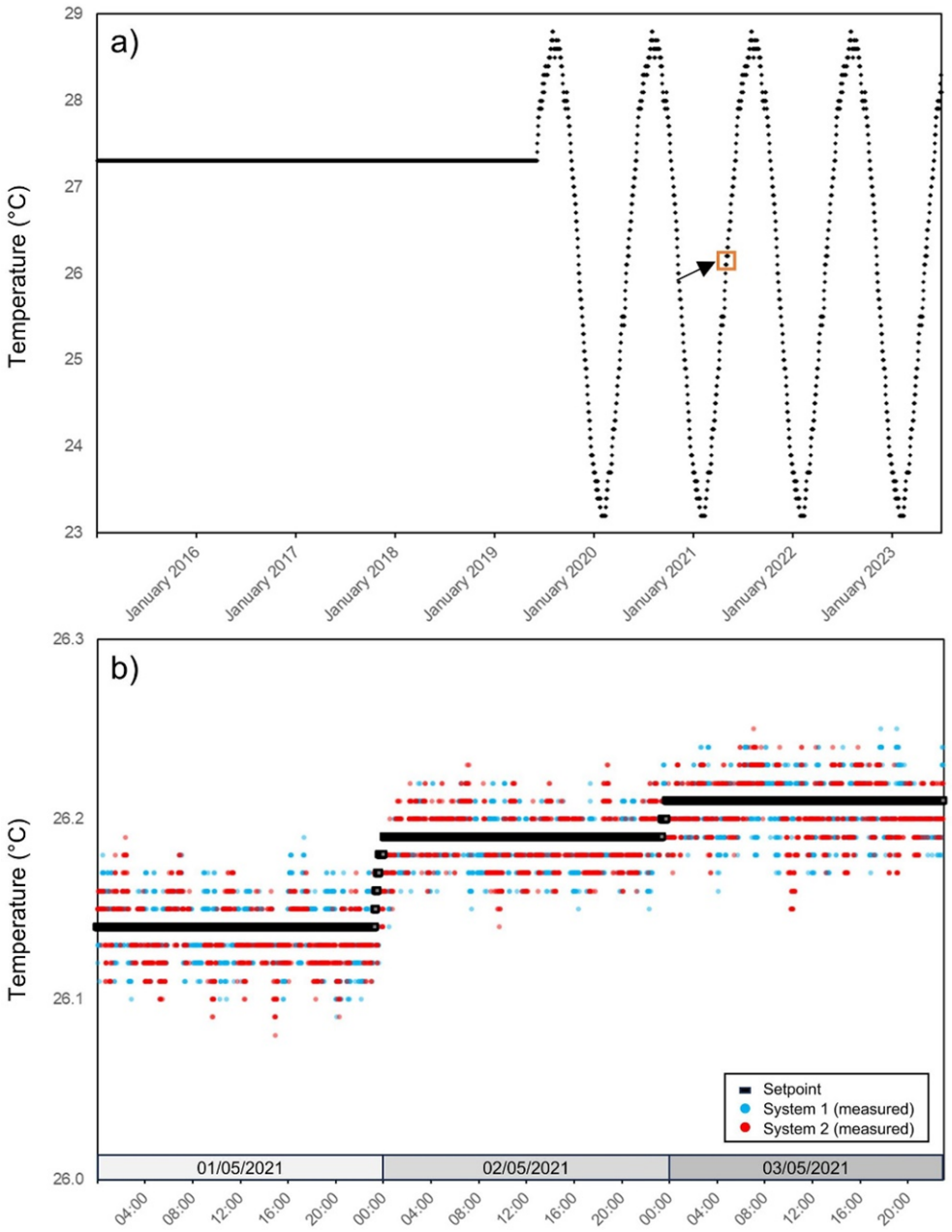


Figure S4. (a) Daily temperature setpoints programmed for coral holding systems between January 2015 and June 2023. See electronic supplementary material, *Data_2_Temperature_entire.csv* for the complete temperature setpoint dataset. (b) Three-day subset (1–3 May 2021) showing measured in-tank temperatures (aggregated to 1-minute intervals) for both systems (red and blue points) compared to programmed setpoints (black points).

2.5 Photosynthetically Active Radiation provided to out-of-season spawning systems

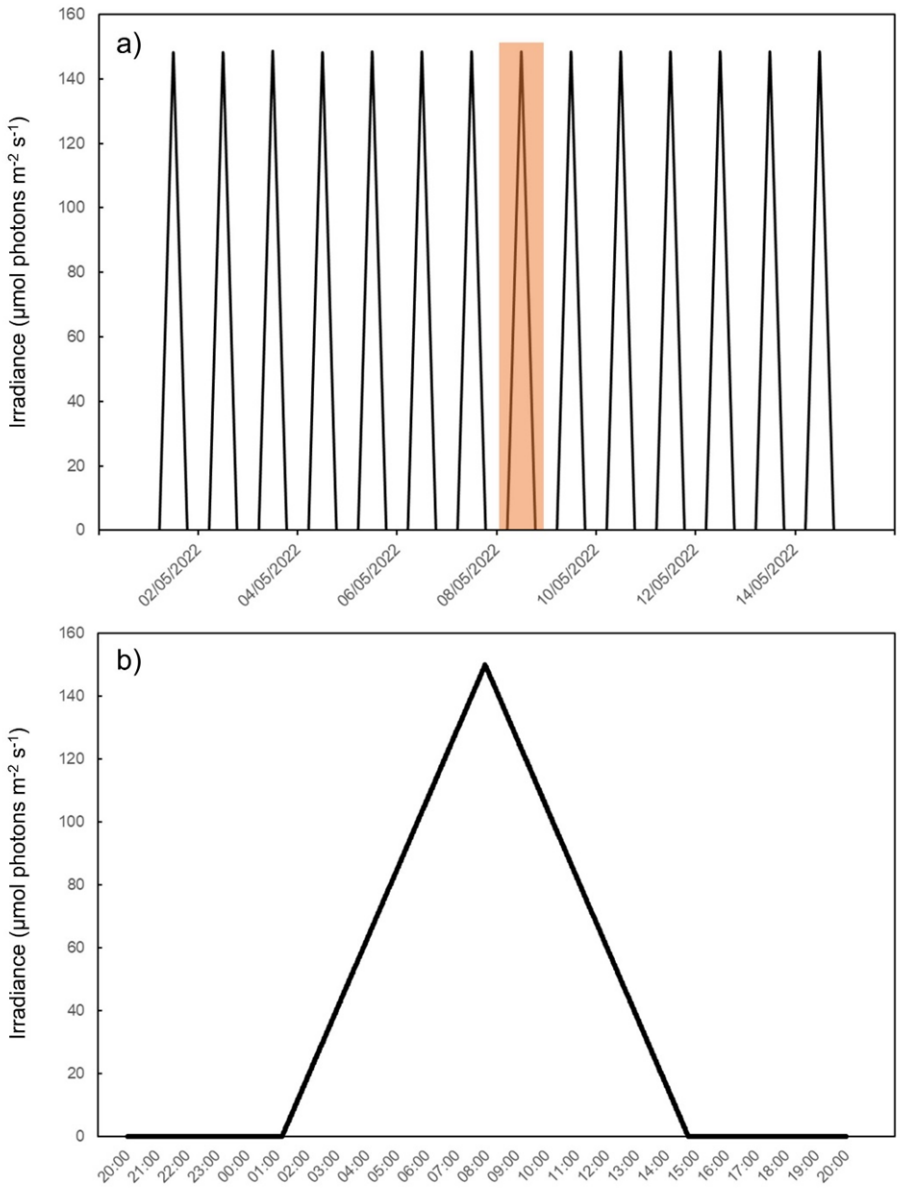


Figure S5. Photosynthetically Active Radiation (PAR; µmol photons m^{-2} s^{-1}) delivered to out-of-season coral holding systems. (a) PAR over a 14-day period (1–14 May 2022). (b) 24-hour irradiance profile during the offset inshore full moon period (7–8 May 2022). Note: A 4-hour phase shift was applied to the photoperiod, resulting in actual lights-on (first light) at 01:08 and lights-off *(last light) at 14:44. These correspond to programmed setpoints of 05:08 and 18:44 in electronic supplementary material, *Data_3_Photoperiod_entire.csv*. Light ramping setpoints for the 24-hour cycle are provided in electronic supplementary material, *Data_4_Photoperiod_24_hours.csv*.

2.6 Moonlight illuminance (lux) provided to out-of-season spawning systems

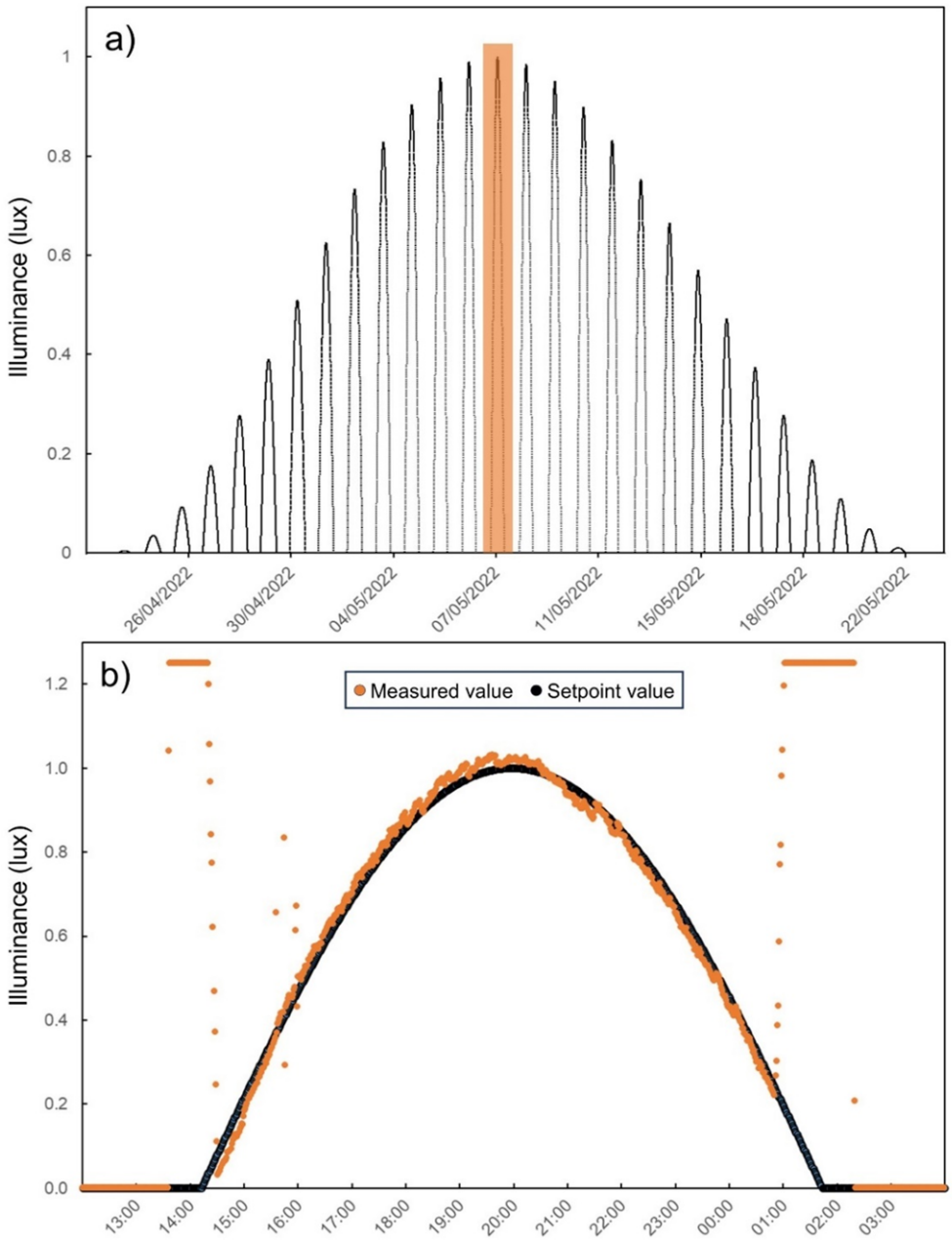


Figure S6. a) Moonlight illuminance (lux) setpoints over a 31-day period (23 April – 23 May 2022) in coral holding tanks, capturing a full lunar cycle. See electronic supplementary material, *Data_5_Moonlight_entire.csv* for moonlight setpoint values (Moonrise, Moonset, illuminance [lux]) provided over the duration of the study. (b) Setpoint (black) and measured (orange) moonlight profiles during the offset inshore full moon period (7–8 May 2022). Spikes in lux values at either end of the cycle result from overlap with daylight photoperiods; transient spikes (15:00–16:00) reflect red headtorch interference during spawning checks. Note: Experimental room is four-hour phase-shifted (bringing Moonrise and Moonset times forward by four hours) from times listed in electronic supplementary material, *Data_5_Moonlight_entire.csv*

2.7 Out-of-season spawning observations

Table S1: Out-of-season spawning observations for F1 coral colonies during the May and June spawning periods of 2022 and 2023. Values are reported as nights after full moon (NAFM) and time after sunset (TAS). * H:M hours:minutes. TAS values were calculated based on the observed time of gamete release relative to the phase shifted sunset times defined in electronic supplementary material, *Data_3_Photoperiod_entire.csv*.

Date	NAFM	UIN	Species	TAS (H:M) *
13/05/2022	6	982000405334969	*Acropora austera*	1:50
14/05/2022	7	982000405335227	*Acropora millepora*	2:35
15/05/2022	8	982000405334834	*Acropora loripes*	2:50
15/05/2022	8	982000405335012	*Acropora loripes*	2:50
15/05/2022	8	982000405335227	*Acropora millepora*	2:50
15/05/2022	8	982000403043086	*Acropora millepora*	2:20
15/05/2022	8	982000405335174	*Acropora elseyi*	2:30
16/05/2022	9	982000405335227	*Acropora millepora*	2:30
16/05/2022	9	982000405334834	*Acropora loripes*	2:50
16/05/2022	9	982000405335012	*Acropora loripes*	2:50
16/05/2022	9	982000404702094	*Acropora elseyi*	2:40
16/05/2022	9	982000404701908	*Acropora elseyi*	2:20
16/05/2022	9	982000405335174	*Acropora elseyi*	2:35
10/06/2022	4	982000404702647	*Acropora loripes*	2:50
10/06/2022	4	982000404702636	*Acropora loripes*	3:05
10/06/2022	4	982000405334783	*Acropora loripes*	2:55
11/06/2022	5	982000364560549	*Acropora loripes*	3:05
11/06/2022	5	982000364560727	*Acropora loripes*	3:05
11/06/2022	5	982000364560477	*Acropora loripes*	3:05
11/06/2022	5	982000405335661	*Acropora loripes*	3:05
12/06/2022	6	982000405334833	*Montipora aequituberculata*	2:05
12/06/2022	6	982000403150965	*Acropora loripes*	3:05
12/06/2022	6	982000405335661	*Acropora loripes*	3:05
12/06/2022	6	982000405334946	*Acropora loripes*	3:10
12/06/2022	6	982000404702712	*Acropora loripes*	3:05
12/06/2022	6	982000405335174	*Acropora elseyi*	2:35
13/06/2022	7	982000403150945	*Acropora loripes*	3:00
13/06/2022	7	982000404702703	*Acropora loripes*	3:05
13/06/2022	7	982000404702647	*Acropora loripes*	3:00
13/06/2022	7	982000405334783	*Acropora loripes*	3:00
13/06/2022	7	982000405335661	*Acropora loripes*	3:00
13/06/2022	7	982000404702727	*Acropora loripes*	3:00
13/06/2022	7	982000404707637	*Acropora loripes*	3:05
13/06/2022	7	982000404701911	*Acropora loripes*	3:05
13/06/2022	7	982000364560547	*Acropora loripes*	3:05
13/06/2022	7	982000405335174	*Acropora elseyi*	2:55
14/06/2022	8	982000404702094	*Acropora elseyi*	2:53
14/06/2022	8	982000405335174	*Acropora elseyi*	2:53
14/06/2022	8	982000405335659	*Acropora elseyi*	2:55
14/06/2022	8	982000405334833	*Montipora aequituberculata*	2:15
14/06/2022	8	982000403150965	*Acropora loripes*	3:00
15/06/2022	9	982000404702720	*Acropora loripes*	3:00
15/06/2022	9	982000404702665	*Acropora loripes*	3:00
15/06/2022	9	982000404702694	*Acropora loripes*	2:55
15/06/2022	9	982000404702712	*Acropora loripes*	3:05
15/06/2022	9	982000403150965	*Acropora loripes*	3:05
15/06/2022	9	982000404702636	*Acropora loripes*	3:05
15/06/2022	9	982000405334783	*Acropora loripes*	3:00
15/06/2022	9	982000364560547	*Acropora loripes*	3:00
15/06/2022	9	982000403150945	*Acropora loripes*	2:55
15/06/2022	9	982000404702703	*Acropora loripes*	3:00

15/06/2022	9	982000404702727	*Acropora loripes*	2:55
15/06/2022	9	982000403150935	*Acropora loripes*	3:00
15/06/2022	9	982000404702637	*Acropora loripes*	3:00
15/06/2022	9	982000405335659	*Acropora elseyi*	2:45
15/06/2022	9	982000404702094	*Acropora elseyi*	2:45
15/06/2022	9	982000405335723	*Acropora hyacinthus*	3:30
15/06/2022	9	982000405334833	*Montipora aequituberculata*	2:05
16/06/2022	10	982000405335227	*Acropora millepora*	2:40
16/06/2022	10	982000403043216	*Acropora millepora*	2:50
16/06/2022	10	982000405334833	*Montipora aequituberculata*	2:05
17/06/2022	11	982000405335007	*Acropora hyacinthus*	3:25
17/06/2022	11	982000405335723	*Acropora hyacinthus*	3:30
17/06/2022	11	982000405334813	*Acropora hyacinthus*	3:40
17/06/2022	11	982000405335659	*Acropora elseyi*	2:55
3/05/2023	2	982000405334775	*Montipora aequituberculata*	1:50
3/05/2023	2	982000405334833	*Montipora aequituberculata*	1:48
3/05/2023	2	982000405335030	*Montipora aequituberculata*	1:50
4/05/2023	3	982000405335024	*Montipora aequituberculata*	1:32
4/05/2023	3	982000405334756	*Montipora aequituberculata*	1:35
4/05/2023	3	982000405334833	*Montipora aequituberculata*	1:35
4/05/2023	3	982000405334775	*Montipora aequituberculata*	1:37
4/05/2023	3	982000405334921	*Montipora aequituberculata*	1:38
4/05/2023	3	982000405335030	*Montipora aequituberculata*	1:40
5/05/2023	4	982000405335030	*Montipora aequituberculata*	1:40
5/05/2023	4	982000405334775	*Montipora aequituberculata*	1:40
5/05/2023	4	982000405334756	*Montipora aequituberculata*	1:40
5/05/2023	4	982000405334833	*Montipora aequituberculata*	1:40
5/05/2023	4	982000405334749	*Acropora austera*	1:42
5/05/2023	4	982000403150975	*Acropora austera*	1:42
6/05/2023	5	982000403043216	*Acropora millepora*	2:10
6/05/2023	5	982000405335227	*Acropora millepora*	2:12
6/05/2023	5	982000404701908	*Acropora elseyi*	2:05
6/05/2023	5	982000405335174	*Acropora elseyi*	2:06
6/05/2023	5	982000405334786	*Acropora austera*	1:20
6/05/2023	5	982000405334794	*Acropora austera*	1:20
6/05/2023	5	982000403150975	*Acropora austera*	1:30
6/05/2023	5	982000405334749	*Acropora austera*	1:20
6/05/2023	5	982000405334768	*Acropora austera*	1:10
6/05/2023	5	982000404701774	*Acropora austera*	1:20
7/05/2023	6	982000403043216	*Acropora millepora*	2:00
31/05/2023	1	982000405334921	*Montipora aequituberculata*	1:33
31/05/2023	1	982000405335024	*Montipora aequituberculata*	1:31
31/05/2023	1	982000405335659	*Acropora elseyi*	2:29
31/05/2023	1	982000404702094	*Acropora elseyi*	2:29
1/06/2023	2	982000405334921	*Montipora aequituberculata*	1:44
1/06/2023	2	982000405335024	*Montipora aequituberculata*	1:46
1/06/2023	2	982000405334833	*Montipora aequituberculata*	1:46
1/06/2023	2	982000365414666	*Acropora millepora*	2:38
1/06/2023	2	982000403043216	*Acropora millepora*	2:39
1/06/2023	2	982000405335173	*Acropora elseyi*	2:34
1/06/2023	2	982000404702094	*Acropora elseyi*	2:35
1/06/2023	2	982000405335174	*Acropora elseyi*	2:34
1/06/2023	2	982000403150933	*Acropora loripes*	2:54
1/06/2023	2	982000403150961	*Acropora loripes*	2:54
1/06/2023	2	982000405334834	*Acropora loripes*	2:56
1/06/2023	2	982000405334946	*Acropora loripes*	2:58
1/06/2023	2	982000403150913	*Acropora loripes*	2:59
1/06/2023	2	982000405334754	*Acropora loripes*	3:00
1/06/2023	2	982000403150901	*Acropora loripes*	2:59
1/06/2023	2	982000403150893	*Acropora loripes*	2:59
1/06/2023	2	982000364560477	*Acropora loripes*	2:59

1/06/2023	2	982000405335007	*Acropora hyacinthus*	3:29
1/06/2023	2	982000405334813	*Acropora hyacinthus*	3:26
1/06/2023	2	982000405335723	*Acropora hyacinthus*	3:29
2/06/2023	3	982000405334756	*Montipora aequituberculata*	1:51
2/06/2023	3	982000405334833	*Montipora aequituberculata*	1:51
2/06/2023	3	982000405335024	*Montipora aequituberculata*	1:52
2/06/2023	3	982000405335030	*Montipora aequituberculata*	1:53
2/06/2023	3	982000405334921	*Montipora aequituberculata*	1:53
3/06/2023	4	982000405335030	*Montipora aequituberculata*	1:53
3/06/2023	4	982000405334833	*Montipora aequituberculata*	1:53
3/06/2023	4	982000405334921	*Montipora aequituberculata*	1:52
4/06/2023	5	982000405335024	*Montipora aequituberculata*	1:49
4/06/2023	5	982000405334833	*Montipora aequituberculata*	1:50
4/06/2023	5	982000405335030	*Montipora aequituberculata*	1:49
4/06/2023	5	982000405335661	*Acropora loripes*	2:56
4/06/2023	5	982000403150965	*Acropora loripes*	2:59
4/06/2023	5	982000404702647	*Acropora loripes*	2:54
5/06/2023	6	982000404702665	*Acropora loripes*	2:54
5/06/2023	6	982000404702720	*Acropora loripes*	2:56
5/06/2023	6	982000403150948	*Acropora loripes*	3:00
5/06/2023	6	982000403150945	*Acropora loripes*	2:59
5/06/2023	6	982000404702703	*Acropora loripes*	2:58
5/06/2023	6	982000404702721	*Acropora loripes*	2:58
5/06/2023	6	982000404702725	*Acropora austera*	1:24
6/06/2023	7	982000403150965	*Acropora loripes*	2:54
6/06/2023	7	982000404702712	*Acropora loripes*	2:48
6/06/2023	7	982000404702647	*Acropora loripes*	2:52
6/06/2023	7	982000403150945	*Acropora loripes*	2:56
6/06/2023	7	982000404702703	*Acropora loripes*	2:59
6/06/2023	7	982000404702637	*Acropora loripes*	2:59
6/06/2023	7	982000404702636	*Acropora loripes*	3:00
6/06/2023	7	982000404702727	*Acropora loripes*	2:58
6/06/2023	7	982000405335661	*Acropora loripes*	2:59
6/06/2023	7	982000364560549	*Acropora loripes*	2:55
6/06/2023	7	982000403150910	*Acropora loripes*	2:59
6/06/2023	7	982000365414682	*Acropora loripes*	2:54
6/06/2023	7	982000405334783	*Acropora loripes*	2:55
6/06/2023	7	982000404702665	*Acropora loripes*	2:54
6/06/2023	7	982000404702721	*Acropora loripes*	3:01
7/06/2023	8	982000404702727	*Acropora loripes*	2:54
7/06/2023	8	982000404702703	*Acropora loripes*	2:48
8/06/2023	9	982000404702637	*Acropora loripes*	2:56
8/06/2023	9	982000404702694	*Acropora loripes*	2:50
8/06/2023	9	982000404702676	*Acropora loripes*	2:52
8/06/2023	9	982000364560547	*Acropora loripes*	2:54
8/06/2023	9	982000404702647	*Acropora loripes*	2:55
8/06/2023	9	982000405334783	*Acropora loripes*	2:57
8/06/2023	9	982000364560549	*Acropora loripes*	2:57
8/06/2023	9	982000404702703	*Acropora loripes*	2:58
8/06/2023	9	982000404702727	*Acropora loripes*	2:59
8/06/2023	9	982000403150945	*Acropora loripes*	3:02
8/06/2023	9	982000404702640	*Acropora loripes*	3:00
8/06/2023	9	982000404702636	*Acropora loripes*	2:52
8/06/2023	9	982000404702665	*Acropora loripes*	2:55
8/06/2023	9	982000365414682	*Acropora loripes*	2:59
8/06/2023	9	982000403150910	*Acropora loripes*	2:57
8/06/2023	9	982000404702720	*Acropora loripes*	2:56
8/06/2023	9	982000404702712	*Acropora loripes*	2:57
8/06/2023	9	982000404701889	*Acropora hyacinthus*	3:19
8/06/2023	9	982000404702655	*Acropora hyacinthus*	3:22

2.8 Out-of-season larval culture fertilisation percentages

Table S2: Fertilisation percentages for larvae cultures established from first filial (F1) coral colonies during out-of-season spawning events in 2022 and 2023. Genotypes refer to individual F1 broodstock colonies.

Year	Date	Species	Genotypes	Replicate 1		Replicate 2		Replicate 3		Average fert. (%)
				Sample size	Embryos	Sample size	Embryos	Sample size	Embryos	
2022										
	15/05/2022	*Acropora millepora*	2	25	21	21	16	23	18	79.7
	16/05/2022	*Acropora elseyi*	3	50	17	57	18	58	18	32.1
	10/06/2022	*Acropora loripes*	3	23	13	28	19	87	58	65.2
	13/06/2022	*Acropora loripes*	9	121	91	43	34	60	47	76.8
2023										
	3/05/2023	*Montipora aequituberculata*	3	46	4	52	5	51	7	10.7
	4/05/2023	*Montipora aequituberculata*	6	108	96	146	133	78	72	90.7
	1/06/2023	*Montipora aequituberculata*	3	104	98	81	77	67	61	93.7
	1/06/2023	*Acropora millepora*	2	89	67	41	32	53	43	77.6
	1/06/2023	*Acropora elseyi*	3	40	31	50	34	108	85	75.8
	1/06/2023	*Acropora loripes*	9	39	33	45	40	112	92	84.2
	1/06/2023	*Acropora hyacinthus*	3	80	65	70	56	114	98	83.0
	8/06/2023	*Acropora loripes*	17	70	66	61	56	66	59	91.9

2.9 Percentage of in-season corals spawned from the literature

Table S3: Summary of literature search results reporting the percentage of colonies spawning under natural (in-season) environmental conditions. Only studies involving Great Barrier Reef (GBR) corals with ≥5 colonies and clearly stated spawning proportions were included. See supplementary methods S1.5 for search criteria and inclusion parameters.

Paper	**Species**					
	Acropora austera	***Acropora elseyi***	***Acropora hyacinthus***	***Acropora loripes***	***Acropora millepora***	***Montipora aequituberculata***
Craggs et al. (2020) [4]					33	
Craggs et al. (2017) [5]			29		100	
Lamb et al. (2024) [6]				68		
Babcock et al. (1986) [11]			100		100	
Schweinsberg et al. (2014) [12]			100			
Average			**76.3**	**68.0**	**77.7**	

2.10 Fertilisation percentages for in-season corals from the literature

Table S4: Summary of literature search results reporting fertilisation success (mean percentage) for coral species spawned under natural (in-season) environmental conditions. Only studies involving Great Barrier Reef (GBR) corals assessed under ambient laboratory conditions were included. See supplementary methods S1.6 for search criteria and inclusion parameters.

Paper	Species					
	Acropora austera	***Acropora elseyi***	***Acropora hyacinthus***	***Acropora loripes***	***Acropora millepora***	***Montipora aequituberculata***
Severati et al. (2024) [13]				95.0		
Lamb et al. (2024) [6]				63.0		
Negri et al. (2005) [14]					96.0	95.0
Willis et al. (1997) [15]		93.0	95.0		49.0	
Chan et al. (2019) [16]			72.0	93.0		
Negri et al. (2007) [17]					90.0	
					98.0	
Iguchi et al. (2007) [18]					92.1	
Ricardo et al. (2018) [19]					85.9	
Hagedorn et al. (2017) [20]					90.0	
Van Oppen et al. (2014) [21]					93.9	
					94.5	
					87.2	
					83.3	
Humphrey et al. (2008) [22]					87.1	
Negri and Heyward (2000) [23]					97.7	
Ricardo et al. (2015) [24]					90.0	
Chua et al. (2013) [25]					88.1	
					75.5	
					84.1	
					86.8	
					99.3	
					100.0	
Markey et al. (2007) [26]					97.0	
					98.0	
Nordborg et al. (2021) [27]					82.0	
Davies (2014) [28]					80.0	
Average		**93.0**	**83.5**	**83.7**	**88.6**	**95.0**

2.11 Settlement assays from out-of-season larval cultures

Table S5: Larval settlement assays results for six coral species collected during out-of-season spawning between May 2022 and June 2023. **Species abbreviations**: A. aus – *Acropora austera*. A. els – *Acropora elseyi*. A. hya – *Acropora hyacinthus*. A. lor – *Acropora loripes*. A. mil – *Acropora millepora*. M. aeq – *Montipora aequituberculata*. **Column abbreviations**: No.Set – Number of larvae that settled. No.NotSet – Number of larvae that did not settle.

Species	SpawnDate	ReadDate	Plate	Well	Treatment	No. Set	No. NotSet	Total	% Set
A. mil	15/05/2022	20/05/2022	1	1	CCA	9	1	10	90.0
A. mil	15/05/2022	20/05/2022	1	2	CCA	8	2	10	80.0
A. mil	15/05/2022	20/05/2022	1	3	CCA	9	1	10	90.0
A. mil	15/05/2022	20/05/2022	2	1	CCA	9	1	10	90.0
A. mil	15/05/2022	20/05/2022	2	2	CCA	9	0	10	90.0
A. mil	15/05/2022	20/05/2022	2	3	CCA	9	1	10	90.0
A. mil	15/05/2022	20/05/2022	2	4	Control	0	10	10	0.0
A. mil	15/05/2022	20/05/2022	2	5	Control	0	9	9	0.0
A. mil	15/05/2022	20/05/2022	2	6	Control	0	10	10	0.0
A. mil	15/05/2022	20/05/2022	2	4	Control	0	11	11	0.0
A. mil	15/05/2022	20/05/2022	2	5	Control	0	10	10	0.0
A. mil	15/05/2022	20/05/2022	2	6	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	1	1	CCA	8	2	10	80.0
A. els	16/05/2022	21/05/2022	1	2	CCA	8	2	10	80.0
A. els	16/05/2022	21/05/2022	1	3	CCA	9	1	10	90.0
A. els	16/05/2022	21/05/2022	2	1	CCA	9	1	10	90.0
A. els	16/05/2022	21/05/2022	2	2	CCA	7	3	10	70.0
A. els	16/05/2022	21/05/2022	2	3	CCA	9	1	10	90.0
A. els	16/05/2022	21/05/2022	2	4	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	2	5	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	2	6	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	2	4	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	2	5	Control	0	10	10	0.0
A. els	16/05/2022	21/05/2022	2	6	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	1	1	CCA	9	1	10	90.0
A. lor	11/06/2022	16/06/2022	1	2	CCA	8	2	10	80.0
A. lor	11/06/2022	16/06/2022	1	3	CCA	8	2	10	80.0
A. lor	11/06/2022	16/06/2022	2	1	CCA	7	3	10	70.0
A. lor	11/06/2022	16/06/2022	2	2	CCA	9	1	10	90.0
A. lor	11/06/2022	16/06/2022	2	3	CCA	7	3	10	70.0
A. lor	11/06/2022	16/06/2022	2	4	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	2	5	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	2	6	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	2	4	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	2	5	Control	0	10	10	0.0
A. lor	11/06/2022	16/06/2022	2	6	Control	0	10	10	0.0
A. lor	15/06/2022	20/06/2022	1	1	CCA	8	2	10	80.0
A. lor	15/06/2022	20/06/2022	1	2	CCA	8	2	10	80.0

A. lor	15/06/2022	20/06/2022	1	3	CCA	9	1	10	90.0
A. lor	15/06/2022	20/06/2022	2	1	CCA	9	1	10	90.0
A. lor	15/06/2022	20/06/2022	2	2	CCA	9	1	10	90.0
A. lor	15/06/2022	20/06/2022	2	3	CCA	5	5	10	50.0
A. lor	15/06/2022	20/06/2022	2	4	Control	0	10	10	0.0
A. lor	15/06/2022	20/06/2022	2	5	Control	0	9	9	0.0
A. lor	15/06/2022	20/06/2022	2	6	Control	0	10	10	0.0
A. lor	15/06/2022	20/06/2022	2	4	Control	0	10	10	0.0
A. lor	15/06/2022	20/06/2022	2	5	Control	0	10	10	0.0
A. lor	15/06/2022	20/06/2022	2	6	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	1	1	CCA	9	1	10	90.0
A. els	14/06/2022	20/06/2022	1	2	CCA	7	3	10	70.0
A. els	14/06/2022	20/06/2022	1	3	CCA	10	0	10	100.0
A. els	14/06/2022	20/06/2022	2	1	CCA	6	4	10	60.0
A. els	14/06/2022	20/06/2022	2	2	CCA	7	3	10	70.0
A. els	14/06/2022	20/06/2022	2	3	CCA	8	2	10	80.0
A. els	14/06/2022	20/06/2022	2	4	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	2	5	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	2	6	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	2	4	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	2	5	Control	0	10	10	0.0
A. els	14/06/2022	20/06/2022	2	6	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	1	1	CCA	8	2	10	80.0
A. mil	16/06/2022	21/06/2022	1	2	CCA	9	1	10	90.0
A. mil	16/06/2022	21/06/2022	1	3	CCA	7	3	10	70.0
A. mil	16/06/2022	21/06/2022	2	1	CCA	8	2	10	80.0
A. mil	16/06/2022	21/06/2022	2	2	CCA	10	0	10	100.0
A. mil	16/06/2022	21/06/2022	2	3	CCA	8	2	10	80.0
A. mil	16/06/2022	21/06/2022	2	4	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	2	5	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	2	6	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	2	4	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	2	5	Control	0	10	10	0.0
A. mil	16/06/2022	21/06/2022	2	6	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	1	1	CCA	9	1	10	90.0
A. hya	17/06/2022	22/06/2022	1	2	CCA	10	0	10	100.0
A. hya	17/06/2022	22/06/2022	1	3	CCA	8	2	10	80.0
A. hya	17/06/2022	22/06/2022	2	1	CCA	8	2	10	80.0
A. hya	17/06/2022	22/06/2022	2	2	CCA	9	1	10	90.0
A. hya	17/06/2022	22/06/2022	2	3	CCA	10	0	10	100.0
A. hya	17/06/2022	22/06/2022	2	4	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	2	5	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	2	6	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	2	4	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	2	5	Control	0	10	10	0.0
A. hya	17/06/2022	22/06/2022	2	6	Control	0	10	10	0.0
M. aeq	4/05/2023	10/05/2023	1	1	CCA	9	1	10	90.0

M. aeq	4/05/2023	10/05/2023	1	2	CCA	7	3	10	70.0
M. aeq	4/05/2023	10/05/2023	1	3	CCA	9	1	10	90.0
M. aeq	4/05/2023	10/05/2023	2	1	CCA	7	3	10	70.0
M. aeq	4/05/2023	10/05/2023	2	2	CCA	9	1	10	90.0
M. aeq	4/05/2023	10/05/2023	2	3	CCA	9	1	10	90.0
M. aeq	4/05/2023	10/05/2023	2	4	Control	0	10	10	0.0
M. aeq	4/05/2023	10/05/2023	2	5	Control	0	10	10	0.0
M. aeq	4/05/2023	10/05/2023	2	6	Control	0	10	10	0.0
M. aeq	4/05/2023	10/05/2023	2	4	Control	0	10	10	0.0
M. aeq	4/05/2023	10/05/2023	2	5	Control	0	9	9	0.0
M. aeq	4/05/2023	10/05/2023	2	6	Control	0	10	10	0.0
A. aus	6/05/2023	11/05/2023	1	1	CCA	8	2	10	80.0
A. aus	6/05/2023	11/05/2023	1	2	CCA	7	3	10	70.0
A. aus	6/05/2023	11/05/2023	1	3	CCA	8	2	10	80.0
A. aus	6/05/2023	11/05/2023	2	1	CCA	9	1	10	90.0
A. aus	6/05/2023	11/05/2023	2	2	CCA	9	1	10	90.0
A. aus	6/05/2023	11/05/2023	2	3	CCA	8	2	10	80.0
A. aus	6/05/2023	11/05/2023	2	4	Control	0	10	10	0.0
A. aus	6/05/2023	11/05/2023	2	5	Control	0	10	10	0.0
A. aus	6/05/2023	11/05/2023	2	6	Control	0	9	9	0.0
A. aus	6/05/2023	11/05/2023	2	4	Control	0	10	10	0.0
A. aus	6/05/2023	11/05/2023	2	5	Control	0	10	10	0.0
A. aus	6/05/2023	11/05/2023	2	6	Control	0	10	10	0.0
A. els	1/06/2023	7/06/2023	1	1	CCA	6	4	10	60.0
A. els	1/06/2023	7/06/2023	1	2	CCA	5	4	9	55.6
A. els	1/06/2023	7/06/2023	1	3	CCA	8	0	8	100.0
A. els	1/06/2023	7/06/2023	2	1	CCA	8	1	9	88.9
A. els	1/06/2023	7/06/2023	2	2	CCA	9	1	10	90.0
A. els	1/06/2023	7/06/2023	2	3	CCA	6	2	8	75.0
A. els	1/06/2023	7/06/2023	2	4	Control	1	10	11	9.1
A. els	1/06/2023	7/06/2023	2	5	Control	0	12	12	0.0
A. els	1/06/2023	7/06/2023	2	6	Control	0	10	10	0.0
A. els	1/06/2023	7/06/2023	2	4	Control	0	10	10	0.0
A. els	1/06/2023	7/06/2023	2	5	Control	0	10	10	0.0
A. els	1/06/2023	7/06/2023	2	6	Control	0	10	10	0.0
A. lor	6/06/2023	11/06/2023	1	1	CCA	8	2	10	80.0
A. lor	6/06/2023	11/06/2023	1	2	CCA	9	1	10	90.0
A. lor	6/06/2023	11/06/2023	1	3	CCA	9	1	10	90.0
A. lor	6/06/2023	11/06/2023	2	1	CCA	10	0	10	100.0
A. lor	6/06/2023	11/06/2023	2	2	CCA	9	1	10	90.0
A. lor	6/06/2023	11/06/2023	2	3	CCA	9	1	10	90.0
A. lor	6/06/2023	11/06/2023	2	4	Control	0	11	11	0.0
A. lor	6/06/2023	11/06/2023	2	5	Control	0	10	10	0.0
A. lor	6/06/2023	11/06/2023	2	6	Control	0	10	10	0.0
A. lor	6/06/2023	11/06/2023	2	4	Control	0	10	10	0.0
A. lor	6/06/2023	11/06/2023	2	5	Control	0	10	10	0.0
A. lor	6/06/2023	11/06/2023	2	6	Control	0	10	10	0.0

2.12 Out-of-season coral oocyte (egg) diameters

Table S6: Oocyte (egg) diameter (µm) measured across six coral species during out-of-season spawning events conducted between May 2022 and June 2023. Values represent mean diameters calculated from unfertilised oocytes ($n \geq 25$ per species) imaged and measured using ImageJ

Date	Species	Vertical egg diameter (µm)	Horizontal egg diameter (µm)	Average egg diameter (µm)
15/05/2022	*Acropora millepora*	562	562	562
15/05/2022	*Acropora millepora*	549	571	560
15/05/2022	*Acropora millepora*	540	548	544
15/05/2022	*Acropora millepora*	572	553	563
15/05/2022	*Acropora millepora*	540	544	542
15/05/2022	*Acropora millepora*	512	507	510
15/05/2022	*Acropora millepora*	576	549	563
15/05/2022	*Acropora millepora*	571	521	546
15/05/2022	*Acropora millepora*	576	548	562
15/05/2022	*Acropora millepora*	526	599	563
15/05/2022	*Acropora millepora*	489	619	554
15/05/2022	*Acropora millepora*	475	558	517
15/05/2022	*Acropora millepora*	530	544	537
15/05/2022	*Acropora millepora*	447	475	461
15/05/2022	*Acropora millepora*	507	531	519
15/05/2022	*Acropora millepora*	558	558	558
15/05/2022	*Acropora millepora*	551	512	532
15/05/2022	*Acropora millepora*	619	553	586
15/05/2022	*Acropora millepora*	557	566	562
15/05/2022	*Acropora millepora*	557	526	542
15/05/2022	*Acropora millepora*	579	548	564
15/05/2022	*Acropora millepora*	566	566	566
15/05/2022	*Acropora millepora*	614	522	568
15/05/2022	*Acropora millepora*	583	557	570
15/05/2022	*Acropora millepora*	561	601	581
15/05/2022	*Acropora millepora*	561	592	577
15/05/2022	*Acropora millepora*	539	522	531
16/05/2022	*Acropora elseyi*	449	432	441
16/05/2022	*Acropora elseyi*	447	462	455
16/05/2022	*Acropora elseyi*	469	456	463
16/05/2022	*Acropora elseyi*	495	482	489
16/05/2022	*Acropora elseyi*	471	462	467
16/05/2022	*Acropora elseyi*	469	458	464
16/05/2022	*Acropora elseyi*	458	456	457
16/05/2022	*Acropora elseyi*	449	456	453
16/05/2022	*Acropora elseyi*	456	438	447
16/05/2022	*Acropora elseyi*	438	443	441
16/05/2022	*Acropora elseyi*	436	423	430
16/05/2022	*Acropora elseyi*	451	443	447
16/05/2022	*Acropora elseyi*	431	458	445
16/05/2022	*Acropora elseyi*	446	446	446
16/05/2022	*Acropora elseyi*	435	449	442
16/05/2022	*Acropora elseyi*	427	446	437
16/05/2022	*Acropora elseyi*	438	459	449
16/05/2022	*Acropora elseyi*	446	457	452
16/05/2022	*Acropora elseyi*	419	443	431
16/05/2022	*Acropora elseyi*	449	449	449

16/05/2022	*Acropora elseyi*	476	465	471
16/05/2022	*Acropora elseyi*	424	443	434
16/05/2022	*Acropora elseyi*	438	446	442
10/06/2022	*Acropora loripes*	555	471	513
10/06/2022	*Acropora loripes*	554	468	511
10/06/2022	*Acropora loripes*	496	502	499
10/06/2022	*Acropora loripes*	488	508	498
10/06/2022	*Acropora loripes*	497	511	504
10/06/2022	*Acropora loripes*	476	471	474
10/06/2022	*Acropora loripes*	547	536	542
10/06/2022	*Acropora loripes*	513	558	536
10/06/2022	*Acropora loripes*	534	582	558
10/06/2022	*Acropora loripes*	509	526	518
10/06/2022	*Acropora loripes*	558	530	544
10/06/2022	*Acropora loripes*	501	586	544
10/06/2022	*Acropora loripes*	505	566	536
10/06/2022	*Acropora loripes*	493	546	520
10/06/2022	*Acropora loripes*	534	518	526
10/06/2022	*Acropora loripes*	550	554	552
10/06/2022	*Acropora loripes*	562	550	556
10/06/2022	*Acropora loripes*	558	550	554
10/06/2022	*Acropora loripes*	526	534	530
10/06/2022	*Acropora loripes*	566	574	570
4/05/2023	*Montipora aequituberculata*	395	390	393
4/05/2023	*Montipora aequituberculata*	408	419	414
4/05/2023	*Montipora aequituberculata*	362	362	362
4/05/2023	*Montipora aequituberculata*	368	387	378
4/05/2023	*Montipora aequituberculata*	416	421	419
4/05/2023	*Montipora aequituberculata*	363	353	358
4/05/2023	*Montipora aequituberculata*	346	354	350
4/05/2023	*Montipora aequituberculata*	354	366	360
4/05/2023	*Montipora aequituberculata*	334	337	336
4/05/2023	*Montipora aequituberculata*	400	392	396
4/05/2023	*Montipora aequituberculata*	378	406	392
4/05/2023	*Montipora aequituberculata*	414	414	414
4/05/2023	*Montipora aequituberculata*	358	350	354
4/05/2023	*Montipora aequituberculata*	421	417	419
4/05/2023	*Montipora aequituberculata*	382	390	386
4/05/2023	*Montipora aequituberculata*	404	409	407
4/05/2023	*Montipora aequituberculata*	323	411	367
4/05/2023	*Montipora aequituberculata*	380	362	371
4/05/2023	*Montipora aequituberculata*	401	404	403
4/05/2023	*Montipora aequituberculata*	372	375	374
4/05/2023	*Montipora aequituberculata*	393	383	388
4/05/2023	*Montipora aequituberculata*	362	417	390
4/05/2023	*Montipora aequituberculata*	396	432	414
4/05/2023	*Montipora aequituberculata*	388	383	386
4/05/2023	*Montipora aequituberculata*	414	406	410
4/05/2023	*Montipora aequituberculata*	378	388	383
4/05/2023	*Montipora aequituberculata*	438	391	415
4/05/2023	*Montipora aequituberculata*	385	391	388
1/06/2023	*Acropora elseyi*	476	470	473
1/06/2023	*Acropora elseyi*	481	475	478
1/06/2023	*Acropora elseyi*	506	513	510
1/06/2023	*Acropora elseyi*	516	504	510
1/06/2023	*Acropora elseyi*	517	527	522

1/06/2023	*Acropora elseyi*	498	486	492
1/06/2023	*Acropora elseyi*	517	525	521
1/06/2023	*Acropora elseyi*	505	517	511
1/06/2023	*Acropora elseyi*	506	511	509
1/06/2023	*Acropora elseyi*	518	508	513
1/06/2023	*Acropora elseyi*	535	512	524
1/06/2023	*Acropora elseyi*	481	489	485
1/06/2023	*Acropora elseyi*	494	512	503
1/06/2023	*Acropora elseyi*	505	512	509
1/06/2023	*Acropora elseyi*	486	486	486
1/06/2023	*Acropora elseyi*	494	499	497
1/06/2023	*Acropora elseyi*	515	502	509
1/06/2023	*Acropora elseyi*	486	484	485
1/06/2023	*Acropora elseyi*	497	486	492
1/06/2023	*Acropora elseyi*	505	507	506
1/06/2023	*Acropora elseyi*	510	502	506
1/06/2023	*Acropora elseyi*	510	503	507
1/06/2023	*Acropora elseyi*	497	502	500
1/06/2023	*Acropora elseyi*	473	461	467
1/06/2023	*Acropora elseyi*	487	487	487
1/06/2023	*Acropora elseyi*	491	484	488
1/06/2023	*Acropora elseyi*	507	505	506
1/06/2023	*Acropora hyacinthus*	602	612	607
1/06/2023	*Acropora hyacinthus*	587	592	590
1/06/2023	*Acropora hyacinthus*	616	636	626
1/06/2023	*Acropora hyacinthus*	608	605	607
1/06/2023	*Acropora hyacinthus*	597	608	603
1/06/2023	*Acropora hyacinthus*	605	624	615
1/06/2023	*Acropora hyacinthus*	593	636	615
1/06/2023	*Acropora hyacinthus*	589	577	583
1/06/2023	*Acropora hyacinthus*	561	561	561
1/06/2023	*Acropora hyacinthus*	602	609	606
1/06/2023	*Acropora hyacinthus*	576	581	579
1/06/2023	*Acropora hyacinthus*	593	601	597
1/06/2023	*Acropora hyacinthus*	655	616	636
1/06/2023	*Acropora hyacinthus*	609	622	616
1/06/2023	*Acropora hyacinthus*	613	596	605
1/06/2023	*Acropora hyacinthus*	564	561	563
1/06/2023	*Acropora hyacinthus*	570	590	580
1/06/2023	*Acropora hyacinthus*	605	593	599
1/06/2023	*Acropora hyacinthus*	605	593	599
1/06/2023	*Acropora hyacinthus*	605	599	602
1/06/2023	*Acropora hyacinthus*	619	608	614
1/06/2023	*Acropora hyacinthus*	578	602	590
1/06/2023	*Acropora hyacinthus*	581	593	587
1/06/2023	*Acropora hyacinthus*	543	605	574
1/06/2023	*Acropora hyacinthus*	579	581	580
1/06/2023	*Acropora hyacinthus*	628	632	630
1/06/2023	*Acropora hyacinthus*	596	633	615
1/06/2023	*Acropora loripes*	485	509	497
1/06/2023	*Acropora loripes*	533	510	522
1/06/2023	*Acropora loripes*	507	526	517
1/06/2023	*Acropora loripes*	520	501	511
1/06/2023	*Acropora loripes*	526	518	522
1/06/2023	*Acropora loripes*	509	518	514
1/06/2023	*Acropora loripes*	552	536	544

1/06/2023	*Acropora loripes*	498	491	495
1/06/2023	*Acropora loripes*	578	514	546
1/06/2023	*Acropora loripes*	507	523	515
1/06/2023	*Acropora loripes*	491	543	517
1/06/2023	*Acropora loripes*	491	471	481
1/06/2023	*Acropora loripes*	503	539	521
1/06/2023	*Acropora loripes*	497	503	500
1/06/2023	*Acropora loripes*	503	491	497
1/06/2023	*Acropora loripes*	491	515	503
1/06/2023	*Acropora loripes*	491	531	511
1/06/2023	*Acropora loripes*	567	499	533
1/06/2023	*Acropora loripes*	531	507	519
1/06/2023	*Acropora loripes*	533	517	525
1/06/2023	*Acropora loripes*	477	479	478
1/06/2023	*Acropora loripes*	503	506	505
1/06/2023	*Acropora loripes*	487	493	490
1/06/2023	*Acropora loripes*	539	511	525
1/06/2023	*Acropora loripes*	509	514	512
1/06/2023	*Acropora loripes*	501	498	500
1/06/2023	*Acropora loripes*	523	517	520
1/06/2023	*Acropora millepora*	543	568	556
1/06/2023	*Acropora millepora*	587	571	579
1/06/2023	*Acropora millepora*	540	559	550
1/06/2023	*Acropora millepora*	581	575	578
1/06/2023	*Acropora millepora*	556	562	559
1/06/2023	*Acropora millepora*	549	552	551
1/06/2023	*Acropora millepora*	571	568	570
1/06/2023	*Acropora millepora*	556	542	549
1/06/2023	*Acropora millepora*	544	528	536
1/06/2023	*Acropora millepora*	544	541	543
1/06/2023	*Acropora millepora*	568	578	573
1/06/2023	*Acropora millepora*	560	556	558
1/06/2023	*Acropora millepora*	560	564	562
1/06/2023	*Acropora millepora*	544	526	535
1/06/2023	*Acropora millepora*	591	575	583
1/06/2023	*Acropora millepora*	578	604	591
1/06/2023	*Acropora millepora*	520	623	572
1/06/2023	*Acropora millepora*	496	496	496
1/06/2023	*Acropora millepora*	566	558	562
1/06/2023	*Acropora millepora*	562	570	566
1/06/2023	*Acropora millepora*	554	566	560
1/06/2023	*Acropora millepora*	557	545	551
1/06/2023	*Acropora millepora*	574	580	577
1/06/2023	*Acropora millepora*	545	551	548
1/06/2023	*Acropora millepora*	586	582	584
1/06/2023	*Acropora millepora*	580	563	572
1/06/2023	*Acropora millepora*	568	568	568
1/06/2023	*Montipora aequituberculata*	384	372	378
1/06/2023	*Montipora aequituberculata*	388	370	379
1/06/2023	*Montipora aequituberculata*	355	346	351
1/06/2023	*Montipora aequituberculata*	340	337	339
1/06/2023	*Montipora aequituberculata*	296	293	295
1/06/2023	*Montipora aequituberculata*	359	357	358
1/06/2023	*Montipora aequituberculata*	315	310	313
1/06/2023	*Montipora aequituberculata*	327	325	326
1/06/2023	*Montipora aequituberculata*	283	282	283

2.13 Physiological traits measured for out-of-season corals against conspecifics reported in the wider literature

Table S7. Comparison of physiological traits for out-of-season spawning (OSS) corals and conspecifics reported in the wider literature (i.e., under natural environmental conditions). For OSS data, values represent the mean, with the observed range shown in superscript parentheses (See electronic supplementary material, tables S2, S5, S6 for raw data). For literature values, the mean is reported, with the superscripted range reflecting the spread of published averages across relevant studies (citations in square brackets). "ND" indicates the trait was not determined under OSS conditions; "–" indicates no comparable value was found in the literature. See electronic supplementary material (sections S1.5–S1.8 and tables S3–S4) for full references and search methodology. *Acropora elseyi* egg diameter values are based on unpublished data from C. Randall (2017, AIMS), supplemented with measurements from Wallace [8].

Species	Colonies spawned (%)		Fertilisation (%)		Settlement (%)		Egg diameter (µm)	
	OSS	Literature [4–6,11,12]	OSS	Literature [6,13–28]	OSS	Literature [7]	OSS	Literature [7–10,29]
Acropora austera	11 (3 - 18)	-	ND	-	82	33	ND	627 (545 – 708)
Acropora elseyi	64 (57 -71)	-	54 (32 -76)	93	80 (78 - 83)	-	475 (450 - 500)	404 (272 – 535)
Acropora hyacinthus	64 (57 -71)	76 (29 -100)	83	84 (72 -95)	90	80	599	520 (484 – 622)
Acropora loripes	79 (75 - 83)	68	80 (65 -92)	84 (63 – 95)	83 (80 – 90)	54	520 (512 - 529)	543 (525 – 560)
Acropora millepora	75	78 (33 -100)	79 (78 -80)	89 (49 – 100)	86 (83 – 88)	77 (73 – 81)	555 (549 - 560)	582 (540 – 667)
Montipora aequituberculata	59 (17 -100)	-	65 (11 -94)	95	83	27 (12 – 42)	361 (336 - 386)	310 (241 – 379)

2.14 Results of NAFM/TAS statistical analysis

Table S8: Average nights after full moon (NAFM) for each coral species across three datasets: out-of-season spawning in 2022 and 2023, and wild spawning observations from Baird et al. [30]. Values represent the mean NAFM calculated per species per dataset.

Species	Year	Avg NAFM
Acropora austera	2022	6
Acropora austera	2023	4.89
Acropora austera	Baird	4
Acropora elseyi	2022	8.42
Acropora elseyi	2023	2.57
Acropora elseyi	Baird	5
Acropora hyacinthus	2022	10.5
Acropora hyacinthus	2023	4.8
Acropora hyacinthus	Baird	4.67
Acropora loripes	2022	7.29
Acropora loripes	2023	6.43
Acropora loripes	Baird	5.67
Acropora millepora	2022	8.67
Acropora millepora	2023	3.43
Acropora millepora	Baird	4.18
Montipora aequituberculata	2022	8.25
Montipora aequituberculata	2023	3.10
Montipora aequituberculata	Baird	4.33

Table S9: Generalised Linear Model (GLM) results for raw NAFM values in each species. The model used a Poisson distribution with dataset (2022, 2023, Baird) as a fixed effect. Adjusted p-values are Bonferroni-corrected for 15 comparisons.

Species	Groups	Estimate	Std. Error	z value	Pr (>|z|)	Null deviance	Residual deviance	AIC	Adj. p (15)
Acropora austera	Intercept	1.6094	0.2582	6.2333	4.57E-10	2.7289, 12 df	2.1387, 10 df	54.9390	**6.85E-09**
	2022	0.3365	0.4577	0.7351	0.4623				1.000
	2023	0.1636	0.2925	0.5595	0.5758				1.000
Acropora elseyi	Intercept	1.7918	0.1361	13.1667	1.36E-39	37.0240 df = 27	12.3760 df = 25	121.4500	**2.05E-38**
	2022	0.4507	0.1654	2.7245	0.0064				0.0966
	2023	-0.5188	0.2419	-2.1446	0.0320				0.4798
Acropora hyacinthus	Intercept	1.7356	0.0551	31.4813	1.57E-217	87.0090 df = 66	70.0750 df = 64	311.9700	**2.35E-216**
	2022	0.7067	0.1574	4.4897	7.13E-06				**0.0001**
	2023	0.0222	0.1937	0.1148	0.9086				1.0000
Acropora loripes	Intercept	1.8971	0.1291	14.6950	6.94E-49	71.5640 df = 100	68.0540 df = 98	462.5400	**1.04E-47**
	2022	0.2179	0.1409	1.5467	0.1219				1.0000
	2023	0.1079	0.1384	0.7792	0.4359				1.0000
Acropora millepora	Intercept	1.6446	0.0328	50.0725	0.0000	280.1500 df = 191	261.2800 df = 189	909.3700	**0.0000**
	2022	0.6241	0.1354	4.6111	4.01E-06				**6.01E-05**
	2023	-0.1565	0.1826	-0.8571	0.3914				1.0000
Montipora aequituberculata	Intercept	1.6740	0.1768	9.4694	2.81E-21	27.4740 df = 38	11.1050 df = 36	148.8900	**4.22E-20**
	2022	0.5506	0.2414	2.2810	0.0225				0.3382
	2023	-0.2621	0.1991	-1.3165	0.1880				1.0000

Table S10. Pairwise comparisons of NAFM between datasets (2022, 2023, Baird) within each species, following significant GLMs. Adjusted p-values are corrected for three pairwise tests per species.

Species	Contrast	Estimate	Std. Error	df	Z ratio	P	Adj. P (3)
Acropora austera	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na
Acropora elseyi	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na
Acropora hyacinthus	Baird.2021 – 2022	-0.7067	0.1574	Inf	-4.4897	2.14E-05	**6.42E-05**
	Baird.2021 – 2023	-0.0222	0.1937	Inf	-0.1148	1.0000	1.0000
	2022 - 2023	0.6845	0.2371	Inf	2.8869	0.0117	**0.0350**
Acropora loripes	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na
Acropora millepora	Baird.2021 – 2022	-0.6241	0.1354	Inf	-4.6111	1.20E-05	**3.61E-05**
	Baird.2021 – 2023	0.1565	0.1826	Inf	0.8571	1.0000	1.0000
	2022 - 2023	0.7806	0.2225	Inf	3.5086	0.0014	**0.0041**
Montipora aequituberculata	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na

Table S11. Average time after sunset (TAS, in minutes) for each coral species across three datasets: out-of-season spawning in 2022 and 2023, and wild spawning observations from Baird et al. (2021). Values represent the mean TAS calculated per species per dataset.

Species	Year	Avg TAS
Acropora austera	2022	110.00
Acropora austera	2023	85.33
Acropora austera	Baird	88.50
Acropora elseyi	2022	164.18
Acropora elseyi	2023	134.43
Acropora elseyi	Baird	174.71
Acropora hyacinthus	2022	211.67
Acropora hyacinthus	2023	205.00
Acropora hyacinthus	Baird	194.03
Acropora loripes	2022	180.39
Acropora loripes	2023	176.40
Acropora loripes	Baird	140.80
Acropora millepora	2022	157.50
Acropora millepora	2023	138.14
Acropora millepora	Baird	157.18
Montipora aequituberculata	2022	127.50
Montipora aequituberculata	2023	104.52
Montipora aequituberculata	Baird	68.50

Table S12. Generalised Linear Model (GLM) results for raw TAS values in each species. The model used a Poisson distribution with dataset (2022, 2023, Baird) as a fixed effect. Adjusted p-values are Bonferroni-corrected for 15 comparisons.

Species	Groups	Estimate	Std. Error	z value	Pr (>|z|)	Null deviance	Residual deviance	AIC	Adj. P (15)
Acropora austera	Intercept	4.4830	0.0752	59.6425	0.0000	18.569, 11 df	12.738, 9 df	94.427	**0.0000**
	2022	0.2175	0.1214	1.7913	0.0733				1.0000
	2023	-0.0364	0.0834	-0.4370	0.6621				1.0000
Acropora elseyi	Intercept	5.1632	0.0286	180.5628	0.0000	182.99 df = 24	142.56 df = 22	320.63	**0.0000**
	2022	-0.0622	0.0370	-1.6790	0.0931				1.0000
	2023	-0.2621	0.0434	-6.0447	1.5e-09				**2.25e-08**
Acropora hyacinthus	Intercept	5.2680	0.0116	452.3425	0.0000	505.20 df = 45	498.77 df = 43	830.76	0.0000
	2022	0.0870	0.0414	2.1041	0.0354				0.5310
	2023	0.0550	0.0333	1.6504	0.0989				1.0000
Acropora loripes	Intercept	4.9473	0.0377	131.2681	0.0000	165.60 df = 94	123.61 df = 92	794.94	**0.0000**
	2022	0.2478	0.0396	6.2614	3.82e-10				**5.73e-09**
	2023	0.2254	0.0391	5.7644	8.20E-09				**1.23E-07**
Acropora millepora	Intercept	5.0574	0.0077	658.9148	0.0000	1168.9 df = 120	1153.0 df = 118	1988.6	**0.0000**
	2022	0.0021	0.0334	0.0616	0.9509				1.0000
	2023	-0.1291	0.0331	-3.9042	9.45e-05				**0.0014**
Montipora aequituberculata	Intercept	4.2268	0.0854	49.4738	0.0000	62.018 df = 34	14.950 df = 32	247.94	**0.0000**
	2022	0.6213	0.0962	6.4563	1.07e-10				**1.61e-09**
	2023	0.4225	0.0873	4.8373	1.32e-06				**1.98e-05**

Table S13. Pairwise comparisons of TAS between datasets (2022, 2023, Baird) within each species, following significant GLMs. Adjusted p-values are corrected for nine pairwise tests per species.

Species	Contrast	Estimate	Std. Error	df	Z ratio	P	Adj. P (9)
Acropora austera	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na
Acropora elseyi	Baird.2021 – 2022	0.0622	0.0370	Inf	1.6790	0.2794	1.0000
	Baird.2021 – 2023	0.2621	0.0434	Inf	6.0447	4.49e-09	**4.04e-08**
	2022 - 2023	0.1999	0.0402	Inf	4.9731	1.98e-06	**1.78e-05**
Acropora hyacinthus	Baird.2021 – 2022	na	na	na	na	na	na
	Baird.2021 – 2023	na	na	na	na	na	na
	2022 - 2023	na	na	na	na	na	na
Acropora loripes	Baird.2021 – 2022	-0.2478	0.0396	Inf	-6.2614	1.14e-09	**1.03e-08**
	Baird.2021 – 2023	-0.2254	0.0391	Inf	-5.7644	2.46e-08	**2.21e-07**
	2022 - 2023	0.0224	0.0160	Inf	1.4012	0.4834	1.0000
Acropora millepora	Baird.2021 – 2022	-0.0021	0.0334	Inf	-0.0616	1.0000	1.0000
	Baird.2021 – 2023	0.1291	0.0331	Inf	3.9042	2.84e-04	**0.0026**
	2022 - 2023	0.1311	0.0457	Inf	2.8669	0.0124	0.1119
Montipora aequituberculata	Baird.2021 – 2022	-0.6213	0.0962	Inf	-6.4563	3.22e-10	**2.90e-09**
	Baird.2021 – 2023	-0.4225	0.0873	Inf	-4.8373	3.95e-06	**3.55e-05**
	2022 - 2023	0.1988	0.0479	Inf	4.1529	9.85e-05	**8.86e-04**

3. Supplementary Results

3.1 Comparative analysis of coral reproductive performance between OSS and literature benchmarks

Coral reproductive metrics observed under the OSS project generally aligned well with, or exceeded, values reported in the wider literature, with a few exceptions. Spawning success varied among species, with *Acropora loripes* (79%) and *A. millepora* (75%) showing the highest proportions of colonies spawned in OSS, comparable to or slightly lower than literature values (e.g., 78% for *A. millepora*) (figure 2, table S1, table S3, and table S7). *A. hyacinthus* showed similar spawning proportions to the literature (64% vs. 76%), while in contrast, *A. austera* exhibited markedly reduced spawning success (11%) relative to other species, for which no literature comparison was available (table S7).

Fertilisation success of corals in the OSS program was broadly consistent with published ranges (table S2, table S4, and table S7). For example, *A. hyacinthus* and *A. loripes* demonstrated fertilisation rates of 83% and 80% respectively, closely matching literature values (both 84%). Average fertilisation success for *A. millepora* in the OSS project (79%) fell slightly below the literature average (89%) but comfortably within the range of reported values (49 – 100%). However, *A. elseyi* and *M. aequituberculata* showed notably reduced fertilisation rates (54% and 65%, respectively) compared to literature means of 93% and 95% (table S7). We highlight however that while fertilisation success for *Montipora aequituberculata* averaged 65%, this was skewed by a single low value (11%) from one replicate. The two remaining fertilisation values (91% and 94%) closely align with the literature-reported maxima (95%), indicating that the lower mean likely reflects an outlier rather than systemic underperformance of this species in the OSS program. Fertilisation data could not be found in the wider literature for *A. austera*.

Larval settlement in OSS was consistently high for most species, exceeding literature benchmarks in all species for which comparisons against literature values were possible (table S5 and table S7). Notably, *A. austera* demonstrated a much higher settlement rate of 82% compared to 33% in the literature, and *M.*

aequituberculata reached 83% settlement which was notably higher than the average of 27% observed in previous studies (table S7). These results suggest strong post-fertilisation viability of corals grown under OSS environmental protocols.

Egg diameter measurements in OSS fell within or exceeded the range of literature values (table S6- S7). *A. elseyi*, *A. hyacinthus* and *M. aequituberculata* in our OOS study exhibited a slight increase in average egg size relative to average literature values, while *A. loripes* and *A. millepora* average egg diameters were slightly smaller than mean values reported for conspecifics under natural spawning windows (table S7).

4. Descriptions of accompanying electronic supplementary data files

Electronic supplementary data files [31] are available at https://doi.org/10.5061/dryad.v15dv4281

Data_1_Temperature_profiles.csv: Average temperature profile (1998-2015) for Davies Reef and the 6-month offset temperature profile used in the study. Source: AIMS weather station at Davies Reef (18.832° S, 147.635° E), recorded at 4 m depth.

Data_2_Temperature_entire.csv: Daily temperature setpoint values applied to out-of-season spawning coral systems over the duration of the study

Data_3_Photoperiod_entire.csv: Photoperiod data used to simulate natural dawn and dusk timings, including civil twilight periods, for the out-of-season spawning system. Includes first light and last light times for Townsville, Australia, offset by 6 month**s.**

Data_4_Photoperiod_24_hours.csv: Setpoint values for daily light irradiance (Photosynthetically Active Radiation [PAR]; µmol photons m^{-2} s^{-1}) delivered to coral holding systems over a 24-hour period during the inshore full moon (20:00, 7 May 2022 – 19:59, 8 May 2022), as shown in figure S5. First light and last light times were derived from *Data 3 – Photoperiod_entire.cs*v.

Data_5_Moonlight_entire.csv: Moonlight settings applied to coral holding tanks, including moonrise and moonset times and illuminance (lux) over the duration of the study. Note: Moonrise and moonset times are phase-shifted by four hours (i.e., advanced by four hours) relative to those listed in the file.

***Data_6_NAFM_TAS_rawdata.xlsx*:** Raw spawning time data for all coral species used in the out-of-season spawning (OSS) study. Includes individual records for nights after full moon (NAFM) and time after sunset (TAS) for each spawning event across 2022 and 2023. Dataset supports statistical comparisons of temporal spawning synchrony across years and against published wild observations (see Baird et al. [30]).

Data_7_NAFM_Rdatatable.csv: Formatted input dataset for statistical analysis of nights after full moon (NAFM), used in the Poisson GLM models described in the study. Includes variables for species, dataset (2022, 2023, Baird), and transformed NAFM values (with +1 offset to meet GLM distributional assumptions). Corresponds to results in supplementary tables S8-10.

Data_8_TAS_Rdatatable.csv: Formatted input dataset for statistical analysis of time after sunset (TAS), used in the Poisson GLM models described in the study. Includes variables for species, dataset (2022, 2023, Baird), and TAS values in minutes. Corresponds to results in supplementary tables S9-S11.

R_Code_Spawning_Synchrony_Analysis: R script used to analyse coral spawning timing (nights after full moon [NAFM] and time after sunset [TAS]) across datasets from 2022, 2023, and published wild observations (Baird et al. [30]). The code performs data wrangling, assumption checks, and Poisson GLM modelling for each species separately, with post-hoc comparisons using estimated marginal means (emmeans). Outputs include descriptive statistics, model summaries, and pairwise comparisons. Associated input/output files: *Data_6_NAFM_TAS_rawdata.xlsx*, *Data_7_NAFM_Rdatatable.csv,* and *Data_8_TAS_Rdatatable.csv*.

5. Supplementary References

1. Fogarty ND, Marhaver KL. 2019 Coral spawning, unsynchronized. *Science (1979)* **365**, 987–988. (doi:10.1126/science.aay7457)

2. LaJeunesse TC, Parkinson JE, Gabrielson PW, Jeong HJ, Reimer JD, Voolstra CR, Santos SR. 2018 Systematic Revision of Symbiodiniaceae Highlights the Antiquity and Diversity of Coral Endosymbionts. *Current Biology* **28**, 2570-2580.e6. (doi:10.1016/j.cub.2018.07.008)

3. Butler CC *et al.* 2023 Formal recognition of host-generalist species of dinoflagellate (Cladocopium, Symbiodiniaceae) mutualistic with Indo-Pacific reef corals. *J Phycol* **59**, 698–711. (doi:10.1111/jpy.13340)

4. Craggs J, Guest J, Davis M, Sweet M. 2020 Completing the life cycle of a broadcast spawning coral in a closed mesocosm. *Invertebr Reprod Dev* **64**, 244–247. (doi:10.1080/07924259.2020.1759704)

5. Craggs J, Guest JR, Davis M, Simmons J, Dashti E, Sweet M. 2017 Inducing broadcast coral spawning ex situ: Closed system mesocosm design and husbandry protocol. *Ecol Evol* **7**, 11066–11078. (doi:10.1002/ece3.3538)

6. Lamb AM *et al.* 2024 Fertile Hybrids Could Aid Coral Adaptation. *Ecol Evol* **14**, e70570. (doi:10.1002/ECE3.70570)

7. Randall CJ, Giuliano C, Stephenson B, Whitman TN, Page CA, Treml EA, Logan M, Negri AP. 2024 Larval precompetency and settlement behaviour in 25 Indo-Pacific coral species. *Commun Biol* **7**, 1–15. (doi:10.1038/s42003-024-05824-3)

8. Wallace CC. 1999 *Staghorn Corals of the World: a Revision of the Coral Genus Acropora (Scleractinia; Astrocoeniina; Acroporidae) Worldwide, with Emphasis on Morphology, Phylogeny and Biogeography*. CSIRO Publishing. See https://books.google.com/books/about/Staghorn_Corals_of_the_World.html?id=Y5uVU4MfIKAC .

9. Madin JS *et al.* 2016 The Coral Trait Database, a curated database of trait information for coral species from the global oceans. *Sci Data* **3**, 1–22. (doi:10.1038/sdata.2016.17)

10. Shlesinger T, Loya Y. 2019 Breakdown in spawning synchrony: A silent threat to coral persistence. *Science (1979)* **365**, 1002–1007. (doi:10.1126/science.aax0110)

11. Babcock RC, Bull GD, Harrison PL, Heyward AJ, Oliver JK, Wallace CC, Willis BL. 1986 Synchronous spawnings of 105 scleractinian coral species on the Great Barrier Reef. *Mar Biol* **90**, 379–394. (doi:10.1007/BF00428562)

12. Schweinsberg M, González Pech RA, Tollrian R, Lampert KP. 2014 Transfer of intracolonial genetic variability through gametes in Acropora hyacinthus corals. *Coral Reefs* **33**, 77–87. (doi:10.1007/s00338-013-1102-5)

13. Severati A, Nordborg FM, Heyward A, Abdul Wahab MA, Brunner CA, Montalvo-Proano J, Negri AP. 2024 The AutoSpawner system - Automated ex situ spawning and fertilisation of corals for reef restoration. *J Environ Manage* **366**, 121886. (doi:10.1016/j.jenvman.2024.121886)

14. Negri A, Vollhardt C, Humphrey C, Heyward A, Jones R, Eaglesham G, Fabricius K. 2005 Effects of the herbicide diuron on the early life history stages of coral. *Mar Pollut Bull* **51**, 370–383. (doi:10.1016/j.marpolbul.2004.10.053)

15. Willis BL, Babcock RC, Harrison PL, Wallace CC. 1997 Experimental hybridization and breeding incompatibilities within the mating systems of mass spawning reef corals. *Coral Reefs* **16**, S53–S65. (doi:10.1007/s003380050242)

16. Chan WY, Peplow LM, van Oppen MJH. 2019 Interspecific gamete compatibility and hybrid larval fitness in reef-building corals: Implications for coral reef restoration. *Sci Rep* **9**, 1–13. (doi:10.1038/s41598-019-41190-5)

17. Negri AP, Marshall PA, Heyward AJ. 2007 Differing effects of thermal stress on coral fertilization and early embryogenesis in four Indo Pacific species. *Coral Reefs* **26**, 759–763. (doi:10.1007/s00338-007-0258-2)

18. Iguchi A, Márquez LM, Knack B, Shinzato C, Van Oppen MJH, Willis BL, Hardie K, Catmull J, Miller DJ. 2007 Apparent involvement of a β1 type integrin in coral fertilization. *Marine Biotechnology* **9**, 760–765. (doi:10.1007/s10126-007-9026-0)

19. Ricardo GF, Jones RJ, Clode PL, Humanes A, Giofre N, Negri AP. 2018 Sediment characteristics influence the fertilisation success of the corals Acropora tenuis and Acropora millepora. *Mar Pollut Bull* **135**, 941–953. (doi:10.1016/j.marpolbul.2018.08.001)

20. Hagedorn M, Carter VL, Henley EM, Van Oppen MJH, Hobbs R, Spindler RE. 2017 Producing Coral Offspring with Cryopreserved Sperm: A Tool for Coral Reef Restoration. *Sci Rep* **7**, 1–9. (doi:10.1038/s41598-017-14644-x)

21. van Oppen MJH, Puill-Stephan E, Lundgren P, De'ath G, Bay LK. 2014 First-generation fitness consequences of interpopulational hybridisation in a Great Barrier Reef coral and its implications for assisted migration management. *Coral Reefs* **33**, 607–611. (doi:10.1007/s00338-014-1145-2)

22. Humphrey C, Weber M, Lott C, Cooper T, Fabricius K. 2008 Effects of suspended sediments, dissolved inorganic nutrients and salinity on fertilisation and embryo development in the coral Acropora millepora (Ehrenberg, 1834). *Coral Reefs* **27**, 837–850. (doi:10.1007/s00338-008-0408-1)

23. Negri AP, Heyward AJ. 2000 Inhibition of Fertilization and Larval Metamorphosis of the Coral Acropora millepora (Ehrenberg, 1834) by Petroleum Products. *Mar Pollut Bull* **41**, 420–427. (doi:10.1016/S0025-326X(00)00139-9)

24. Ricardo GF, Jones RJ, Clode PL, Humanes A, Negri AP. 2015 Suspended sediments limit coral sperm availability. *Sci Rep* **5**, 18084. (doi:10.1038/srep18084)

25. Chua CM, Leggat W, Moya A, Baird AH. 2013 Temperature affects the early life history stages of corals more than near future ocean acidification. *Mar Ecol Prog Ser* **475**, 85–92. (doi:10.3354/MEPS10077)

26. Markey KL, Baird AH, Humphrey C, Negri AP. 2007 Insecticides and a fungicide affect multiple coral life stages. *Mar Ecol Prog Ser* **330**, 127–137. (doi:10.3354/MEPS330127)

27. Nordborg FM, Brinkman DL, Ricardo GF, Agustí S, Negri AP. 2021 Comparative sensitivity of the early life stages of a coral to heavy fuel oil and UV radiation. *Science of The Total Environment* **781**, 146676. (doi:10.1016/j.scitotenv.2021.146676)

28. Davies SW. 2014 Understanding coral dispersal. University of Texas, Austin. See https://repositories.lib.utexas.edu/items/826aa54c-34b5-44ef-9092-38efd5a927e4.

29. Babcock RC, Heyward AJ. 1986 Larval development of certain gamete-spawning scleractinian corals. *Springer* **5**, 111–116. (doi:10.1007/BF00298178)

30. Baird A, Guest J, Edwards A, Bauman A, Bouwmeester J, Mera H, et al. 2021 An Indo-Pacific coral spawning database. *Sci Data* **8**, 35. (doi:10.1038/s41597-020-00793-8)

31. Koukoumaftsis LP, Salmon M, Everson G, Hughes DJ, Varkey D, Heyward AJ, Severati A, Humphrey C. 2025 Data from: Controlled out-of-season spawning of reef-forming corals using offset environmental cues. Dryad Digital Repository https://doi.org/10.5061/dryad.v15dv4281